

CodeGraph “ Boba Code-Intelligence Integration

Revision: 1 **Last modified:** 2026-06-06T14:40:00Z **Authority:** Constitution §11.4.78 (CodeGraph mandate) · §11.4.79 (own-org submodules included, third-party excluded) · §11.4.80 (update/sync automation) · §11.4.10 (secrets never indexed) · §11.4.77 (regeneration mechanism)

CodeGraph (@colbymchenry/codegraph, <https://github.com/colbymchenry/codegraph>) is a 100%-local SQLite semantic knowledge graph of this codebase, exposed to AI coding agents over MCP. No cloud, no external API. It gives agents instant, consistent symbol / caller / callee / impact resolution instead of repeated shallow file scanning.

Install (§11.4.78)

```
npm install -g @colbymchenry/codegraph      # Node 18+, NO sudo (user-writable)
codegraph --version                          # confirm the binary runs (executable)
```

On this host the binary resolves to /opt/homebrew/bin/codegraph (the npm-managed shim). A standalone-installer shim (~/.local/bin/codegraph) must NOT shadow it “ if codegraph --version reports an older version than npm view @colbymchenry/codegraph version, a stale shim is winning on PATH; remove it so the npm-managed binary is active.

Configuration model “ zero-config (important)

CodeGraph 0.9.x is **zero-config**: there is **no** .codegraph/config.json **exclude list**. What gets indexed is decided by:

- 1. **Built-in defaults** “ dependency/build/cache dirs (node_modules, vendor, dist, build, target, .venv, Pods, etc) are skipped even with no .gitignore, plus any file larger than **1 MB**.
- 2. **.gitignore** “ honored directly (root + nested). **To keep something out of the index, add it to .gitignore**. To pull a default-excluded dir back in, add a negation (!path/).

(Source: the installed package README.md “ “Configuration“.)

What is tracked vs ignored

Path	Git	CodeGraph index	Why
.codegraph/ codegraph.db (+ *.db-wal/*.db-shm/cache/)	gitignored	n/a	Build artifact; regenerate with codegraph index (§11.4.77)
.mcp.json	tracked	n/a	MCP wiring, committed (§11.4.78)

Path	Git	CodeGraph index	Why
scripts/ codegraph_validate.sh	tracked	indexed	Anti-bluff validator (Â§11.4.78 step 4)
docs/codegraph/ Status.md	tracked	n/a	Append-only event ledger (Â§11.4.80)

Exclusions â€” secrets (Â§11.4.10) + third-party (Â§11.4.79)

Secrets are never indexed. Root .gitignore already excludes .env, .env.*, *.env, .qbit.env, *secrets*, config/qBittorrent/, config/jackett/, config/boba.db*. Because CodeGraph honors .gitignore, none of these reach the index. Audited: 0 real secret paths in the index.

Third-party submodule excluded. submodules/jackett (upstream Jackett/Jackett, C#/.NET) is third-party â€” Â§11.4.79 requires it OUT of the index. CodeGraph 0.9.9 does index C#, so an explicit .gitignore entry submodules/jackett/ was added. This is **inert for git** (the submodule is a tracked gitlink via .gitmodules; an ignore entry on a tracked path does not un-track it) but tells CodeGraph to skip it. Audited: 0 submodules/jackett paths in the index (down from 890 before the exclusion).

Own-org submodule INCLUDED. constitution/ (HelixDevelopment, own-org per Â§11.4.79) is deliberately NOT ignored, so its Go workable-items sources are indexed and resolve cross-submodule (e.g.Â versionTagsCmd).

Index / re-generate

```
codegraph index .          # full (re)index â€” the Â§11.4.77 regeneration
codegraph index . --force  # force a clean re-index
codegraph sync .          # incremental update since last index
codegraph status .        # files / nodes / edges
```

Current index: **509 files, 8906 nodes, 17025 edges** (go, python, typescript, properties, xml, yaml).

MCP wiring (Claude Code, project-scoped + committed)

.mcp.json at the repo root registers the CodeGraph MCP server over **stdio**:

```
{
  "mcpServers": {
    "codegraph": { "command": "codegraph", "args": ["serve", "--mcp"], "env": {} }
  }
}
```

serve --mcp is the stdio transport; the command is the bare codegraph on PATH (no hardcoded host path, per Â§11.4.78). It exposes 8 tools: codegraph_search, codegraph_callers, codegraph_callees, codegraph_impact, codegraph_node, codegraph_explore, codegraph_status, codegraph_files.

Other agents: register the same codegraph serve --mcp stdio server via that agentâ€™s MCP config (Cursor, opencode opencode.json, Qwen .qwen/settings.json, Codex, etc.) â€” codegraph install automates this for supported agents.

Anti-bluff verification (Â§11.4.78 step 4)

```
bash scripts/codegraph_validate.sh          # 7 checks; exit 0 = all PASS
```

The validator proves the index is real and correctly scoped â€” six checks plus an **unforgeable MCP challenge**: it drives the real `codegraph serve --mcp` server and asserts the `codegraph_status` node count returned over MCP equals the CLI node count. An agent answering from its own file-reading tools cannot fabricate that number. The other checks assert: index reality (`nodes > 0`), own-code symbol resolution, own-org constitution/ symbol resolution (Â§11.4.79 included), and zero submodules/ jackett / zero secret paths in the index (Â§11.4.79 + Â§11.4.10 excluded). The path-audit detection logic is falsifiability-rehearsed (a synthetic leaked path is detected) so the green PASS is not a tautology.

Update + sync automation + cadence (Â§11.4.80)

Scripts are **inherited by reference** from the constitution submodule â€” never copied:

```
bash constitution/scripts/codegraph_update.sh      # npm-update to latest
bash constitution/scripts/codegraph_sync.sh .      # status â†‘ sync â†‘ s
```

`codegraph_sync.sh` appends every run to BOTH this projectâ€™s `docs/codegraph/Status.md` and the constitutionâ€™s own ledger. **Cadence**: weekly floor (Â§11.4.45 status-digest cadence); may run more often.

Troubleshooting

- **codegraph reports an old version after npm update** â€” a stale standalone shim on PATH (commonly `~/local/bin/codegraph`) is shadowing the npm-managed binary. Remove the stale shim.
- **A path you expect is missing from the index** â€” check it isnâ€™t `.gitignore`d, default-excluded (`node_modules/dist/`), > 1 MB, or an unsupported language.
- **Third-party code appears in the index** â€” add its path to `.gitignore` (per the zero-config exclude model) and `codegraph index . --force`.